Week - 4

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1. CREATE DATABASE

```sql
DROP DATABASE IF EXISTS bookflow_db;
CREATE DATABASE bookflow_db;
USE bookflow_db;
```

Output

Action Output

	#	Time	Action	Message
✔	1	19:37:24	DROP DATABASE IF EXISTS bookflow_db	5 row(s) affected
✔	2	19:37:26	CREATE DATABASE bookflow_db	1 row(s) affected
✔	3	19:37:27	USE bookflow_db	0 row(s) affected

2. CREATE BOOKS TABLE

```sql
CREATE TABLE Books (
    book_id INT AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(200) NOT NULL,
    author VARCHAR(150) NOT NULL,
    genre VARCHAR(100),
    description TEXT,
    published_year INT
);
```

Output

Action Output

	#	Time	Action	Message
✔	1	19:37:24	DROP DATABASE IF EXISTS bookflow_db	5 row(s) affected
✔	2	19:37:26	CREATE DATABASE bookflow_db	1 row(s) affected
✔	3	19:37:27	USE bookflow_db	0 row(s) affected
✔	4	19:38:29	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NUL...	0 row(s) affected

3. INSERT BOOK DATA

```sql
INSERT INTO Books(title, author, genre, description, published_year) VALUES
(
    'The Martian',
    'Rohit Sai',
    'Science Fiction',
    'An astronaut stranded on Mars must survive using science and engineering.',
    2011
),
(
    'Dune',
    'Frank Herbert',
    'Science Fiction',
    'A young nobleman becomes involved in the politics and destiny of a desert planet.',
    1965
),
(
```

Output

Action Output

	#	Time	Action	Message
✔	1	19:37:24	DROP DATABASE IF EXISTS bookflow_db	5 row(s) affected
✔	2	19:37:26	CREATE DATABASE bookflow_db	1 row(s) affected
✔	3	19:37:27	USE bookflow_db	0 row(s) affected
✔	4	19:38:29	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NUL...	0 row(s) affected
✔	5	19:40:07	INSERT INTO Books(title, author, genre, description, published_year) VALUES ('The Martian', 'Rohit Sai', '...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

4. DISPLAY BOOKS

```sql
SELECT
    book_id,
    title,
    author,
    genre,
    published_year
FROM Books
ORDER BY book_id;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

book_id	title	author	genre	published_year
1	The Martian	Rohit Sai	Science Fiction	2011
2	Dune	Frank Herbert	Science Fiction	1965
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
4	Pride and Prejudice	Jane Austen	Romance	1813
5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
6	Foundation	Isaac Asimov	Science Fiction	1951
7	The Hobbit	J.R. Tolkien	Fantasy	1937
8	Enders Game	Orson Scott Card	Science Fiction	1985
9	Harry Potter and the Sorcerers Stone	J.K. Rowling	Fantasy	1997
10	Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	1870
NULL	NULL	NULL	NULL	NULL

Books 1 ×

Output

Action Output

#	Time	Action	Message
2	19:37:26	CREATE DATABASE bookflow_db	1 row(s) affected
3	19:37:27	USE bookflow_db	0 row(s) affected
4	19:38:29	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
5	19:40:07	INSERT INTO Books(title, author, genre, description, published_year) VALUES ('The Martian', 'Rohit Sai', ...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
6	19:41:59	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0, ...	10 row(s) returned

5. ADD EMBEDDING COLUMN

```sql
ALTER TABLE Books
ADD COLUMN embedding JSON;
```

Output

Action Output

#	Time	Action	Message
3	19:37:27	USE bookflow_db	0 row(s) affected
4	19:38:29	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
5	19:40:07	INSERT INTO Books(title, author, genre, description, published_year) VALUES ('The Martian', 'Rohit Sai', ...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
6	19:41:59	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0, ...	10 row(s) returned
7	20:05:31	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

6. INSERT MOCK VECTORS

```sql
WHERE book_id = 3;
UPDATE Books
SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
WHERE book_id = 4;
UPDATE Books
SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
WHERE book_id = 5;
UPDATE Books
SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
WHERE book_id = 6;
UPDATE Books
SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
WHERE book_id = 7;
UPDATE Books
SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
WHERE book_id = 8;
UPDATE Books
SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)
WHERE book_id = 9;
UPDATE Books
SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)
WHERE book_id = 10;
```

Output

Action Output

#	Time	Action	Message
13	20:07:38	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
14	20:07:40	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
15	20:07:42	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
16	20:07:44	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
17	20:07:46	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

7. VERIFY ALL VECTORS

```sql
SELECT
    book_id,
    title,
    embedding
FROM Books
ORDER BY book_id;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.65, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.88, 0.85, 0.15]
9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]
NULL	NULL	NULL

Books 2 ×

Output

Action Output

#	Time	Action	Message
14	20:07:40	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
15	20:07:42	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
16	20:07:44	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
17	20:07:46	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
18	20:08:55	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned

8. L2 DISTANCE -- Compare every book with Book 1 vector: -- [0.90, 0.80, 0.20]

```sql
SELECT
    book_id,
    title,
    author,
    genre,
    SQRT(
        POW(
            CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[0]')) AS DECIMAL(10,5))
            - 0.90,
            2
        )
        +
        POW(
            CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[1]')) AS DECIMAL(10,5))
            - 0.80,
            2
        )
        +
        POW(
            CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[2]')) AS DECIMAL(10,5))
            - 0.20,
            2
```

```sql
            - 0.20,
            2
        )
    ) AS l2_distance
FROM Books
WHERE embedding IS NOT NULL
ORDER BY l2_distance ASC
LIMIT 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

book_id	title	author	genre	l2_distance
1	The Martian	Rohit Sai	Science Fiction	0
8	Enders Game	Orson Scott Card	Science Fiction	0.07348469228349536
2	Dune	Frank Herbert	Science Fiction	0.08660254037844388

Result 3 ×

Output

Action Output

#	Time	Action	Message
15	20:07:42	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
16	20:07:44	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
17	20:07:46	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
18	20:08:55	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned
19	20:10:07	SELECT book_id, title, author, genre, SQRT(POW(CAST(JSON_UNQUOTE(JSON_...	3 row(s) returned

9. COSINE SIMILARITY -- Compare every book with Book 1 vector: -- [0.90, 0.80, 0.20]

```sql
SELECT
    book_id,
    title,
    author,
    genre,
    (
        (
            CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[0]')) AS DECIMAL(10,5)) * 0.90
            +
            CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[1]')) AS DECIMAL(10,5)) * 0.80
            +
            CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[2]')) AS DECIMAL(10,5)) * 0.20
        )
        /
        (
            SQRT(
                POW(CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[0]')) AS DECIMAL(10,5)), 2)
                +
                POW(CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[1]')) AS DECIMAL(10,5)), 2)
                +
                POW(CAST(JSON_UNQUOTE(JSON_EXTRACT(embedding, '$[2]')) AS DECIMAL(10,5)), 2)
            )
            *
            SQRT(
                POW(0.90, 2)
                +
                POW(0.80, 2)
                +
                POW(0.20, 2)
            )
        )
    ) AS cosine_similarity
FROM Books
WHERE embedding IS NOT NULL
ORDER BY cosine_similarity DESC
LIMIT 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

book_id	title	author	genre	cosine_similarity
1	The Martian	Rohit Sai	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9986169165587373
8	Enders Game	Orson Scott Card	Science Fiction	0.9982532606477712

Result 4 ×

Output

Action Output

#	Time	Action	Message
16	20:07:44	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
17	20:07:46	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
18	20:08:55	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned
19	20:10:07	SELECT book_id, title, author, genre, SQRT(POW(CAST(JSON_UNQUOTE(JSON_...	3 row(s) returned
20	20:12:58	SELECT book_id, title, author, genre, ((CAST(JSON_UNQUOTE(JSON_EXTRACT(e...	3 row(s) returned